Lecture 21: Beyond One Speed (Two-Group Diffusion Theory)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: The Limit of "One-Speed"

Until now, we have assumed all neutrons behave like a single "gas" diffusing through the core. We lumped everything into effective constants (D, Σ_a).
The Reality:

1. Neutrons are born **Fast** (2 MeV).
2. They must slow down to become **Thermal** (0.025 eV).
3. Fast neutrons travel much further (longer Mean Free Path) than Thermal neutrons.
4. Materials like water reflect Fast neutrons differently than Thermal neutrons.

To model a real reactor with varying composition ($k_\infty(r)$), we must track these two populations separately. This is the **Two-Group Approximation**.

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1 The Two "Buckets" of Neutrons

We divide the neutron spectrum into two groups:

- **Group 1 (Fast):** Neutrons with energy $E > E_{thermal}$.
  - **Source:** Fission. (All neutrons are born here).
  - **Loss:** Scattering down to Group 2 ($\Sigma_{1\to 2}$) OR Leakage. (Absorption is usually small compared to scattering, except for resonance capture).
- **Group 2 (Thermal):** Neutrons with energy $E \approx k_B T$.
  - **Source:** Scattering down from Group 1.
  - **Loss:** Absorption (Σ_{a2}) causing fission or capture (only a fraction of absorptions cause fission).

2 The Coupled Equations

Instead of one diffusion equation ($D\nabla^2\phi + \cdots = 0$), we now have a system of two coupled differential equations.

2.1 The Fast Group (Group 1)

$$\underbrace{D_1\nabla^2\phi_1}_{\text{Diffusion}} - \underbrace{\Sigma_{1\to 2}\phi_1}_{\text{Removal to Group 2}} + \underbrace{\frac{1}{k}\nu\Sigma_{f2}\phi_2}_{\text{Source from Fission}} = 0$$

Note the coupling: The source of Fast neutrons comes from Fissions caused by **Thermal** neutrons (ϕ_2). We are ignoring the additional contribution of fissions due to fast neutrons, reasonable due to the huge difference in fission capture cross-sections.

2.2 The Thermal Group (Group 2)

$$\underbrace{D_2\nabla^2\phi_2}_{\text{Diffusion}} - \underbrace{\Sigma_{a2}\phi_2}_{\text{Absorption}} + \underbrace{\Sigma_{1\to 2}\phi_1}_{\text{Source from Group 1}} = 0$$

Note the coupling: The source of Thermal neutrons is the "slowing down" from the **Fast** group (ϕ_1).

3 Characteristic Lengths: How Far Do They Go?

To make sense of the coupled equations, we group the constants into characteristic areas (length squared) that describe how far a neutron travels during its life in that specific group.

3.1 1. Fermi Age (τ)

For the Fast Group, the "diffusion length" before slowing down is historically called the **Fermi Age** (τ), even though it has units of cm^2, not time.

$$\tau = \frac{D_1}{\Sigma_{1\to 2}}$$

This represents one-sixth of the mean square crow-flight distance a neutron travels from the exact moment it is born in fission until it slows down to thermal energies. For light water, $\tau \approx 27\ \text{cm}^2$.

3.2 2. Thermal Diffusion Area (L^2)

For the Thermal Group, the parameter is the standard diffusion area we used in one-speed theory.

$$L^2 = \frac{D_2}{\Sigma_{a2}}$$

This represents the area a thermal neutron covers while bouncing around *after* it has slowed down, right up until it is finally absorbed. For light water, $L^2 \approx 8.1\ \text{cm}^2$.

Physical Takeaway: Because $\tau > L^2$ in water, neutrons travel much further while fast than while thermal. Therefore, most neutrons that leak out of a PWR do so while they are fast!

4 Analytical Solution: The Reflector Hump in a Slab

To see the true power of Two-Group theory, let's solve these equations analytically for the reflector region of a 1D slab reactor.

Assume we have a core ending at $x = a$, and a semi-infinite water reflector extending from $x = a$ to $x = \infty$. To make the math clean, let $z = x - a$ be the distance into the reflector.

In the reflector, there is no fissile material, so $\Sigma_f = 0$. The two-group equations simplify significantly:

4.1 1. The Fast Flux in the Reflector (ϕ_{1R})

Without a fission source, the fast group equation is purely a balance of diffusion and removal (slowing down):

$$D_{1R}\frac{d^2\phi_{1R}}{dz^2} - \Sigma_{1R}\phi_{1R} = 0$$

Dividing by D_{1R}, we define the inverse fast diffusion length $\kappa_1^2 = \frac{\Sigma_{1R}}{D_{1R}} = \frac{1}{\tau_R}$. The solution that doesn't blow up at infinity is a simple exponential decay:

$$\phi_{1R}(z) = Ae^{-\kappa_1 z}$$

Where A is the fast flux at the core boundary ($z = 0$). Fast neutrons simply leak into the water and exponentially slow down.

4.2 2. The Thermal Flux in the Reflector (ϕ_{2R})

The thermal equation is driven by the slowing down of those fast neutrons:

$$D_{2R}\frac{d^2\phi_{2R}}{dz^2} - \Sigma_{aR}\phi_{2R} + \Sigma_{1R}\phi_{1R}(z) = 0$$

Dividing by D_{2R} and defining the inverse thermal diffusion length $\kappa_2^2 = \frac{\Sigma_{aR}}{D_{2R}} = \frac{1}{L_R^2}$, we get a non-homogeneous differential equation:

$$\frac{d^2\phi_{2R}}{dz^2} - \kappa_2^2\phi_{2R} = -\frac{\Sigma_{1R}}{D_{2R}}Ae^{-\kappa_1 z}$$

The general solution is the sum of a homogeneous part (thermal neutrons diffusing and being absorbed) and a particular part (thermal neutrons being continuously "born" from the fast flux):

$$\phi_{2R}(z) = Be^{-\kappa_2 z} + Ce^{-\kappa_1 z}$$

By substituting the particular solution ($Ce^{-\kappa_1 z}$) back into the differential equation, we can solve for the coupling constant C:

$$C = \left(\frac{\Sigma_{1R}/D_{2R}}{\kappa_2^2 - \kappa_1^2}\right)A$$

4.3 The Mathematics of the "Hump"

The physical reality of light water is that fast neutrons travel much further than thermal neutrons ($\tau_R > L_R^2$). Therefore, $\kappa_1 < \kappa_2$. This makes the constant C positive.

However, when we apply the boundary conditions at the core interface ($z = 0$) to match the core fluxes and currents, the constant B (the homogeneous amplitude) evaluates to a **negative** number.

This gives us the exact mathematical equation for the thermal flux in the water:

$$\phi_{2R}(z) = Ce^{-\kappa_1 z} - |B|e^{-\kappa_2 z}$$

Physical Meaning: Because $\kappa_2 > \kappa_1$, the negative term ($|B|e^{-\kappa_2 z}$) decays away very quickly near the boundary. Meanwhile, the positive term ($Ce^{-\kappa_1 z}$) persists much deeper into the water.

Near the boundary, as the negative term rapidly vanishes, the total thermal flux $\phi_{2R}(z)$ actually *increases*, creating a localized maximum. This is the exact mathematical origin of the **Reflector Hump**. Fast leakage neutrons travel a short distance into the water, thermalize, and create a surplus of slow neutrons that stack up before finally diffusing away or absorbing.

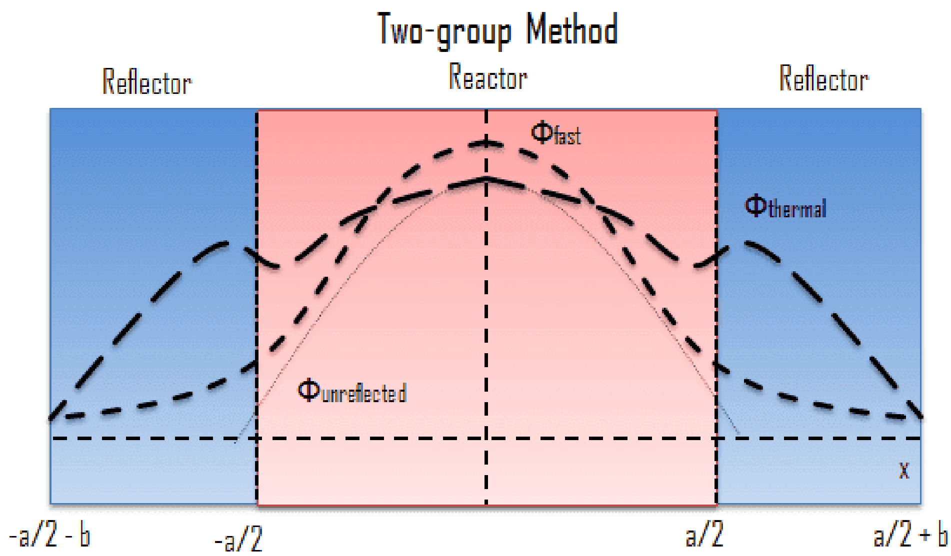

Figure 1: Two-Group neutron flux distribution in a reflected reactor. Note the rise in thermal flux (ϕ_2) in the reflector region, known as the "Reflector Hump," caused by the slowing down of fast leakage neutrons.

5 Redefining Reflector Savings (δ)

Earlier in the semester, we treated the reflector as a simple "extrapolated boundary" by adding a physical distance, the Reflector Savings (δ), to the bare core radius. We treated δ as a given empirical constant.

Two-group theory gives us the rigorous definition of δ. By matching the core flux $\phi_C(x)$ to our new analytical reflector solution $\phi_R(z)$ using the continuity of flux and current at the interface, the requirement that the boundary conditions balance perfectly yields an eigenvalue equation.

Instead of guessing where the flux goes to zero, the core designer calculates δ exactly based on the scattering properties (Σ_{1R}) and diffusion lengths (κ_1, κ_2) of the reflector material. A better reflector slows down more fast neutrons near the boundary, increasing the hump, pushing the effective boundary further out, and explicitly increasing δ.

6 Computational Reality

While we solve 2-group problems by hand for simple slabs, real industry calculations use computer codes (like MCNP or CASMO/SIMULATE) that use:

- **Groups:** 2 (LWR global), 7 (CANDU), or even thousands (Spectrum codes).
- **Mesh:** The reactor is divided into thousands of nodes (x, y, z).
- **Iteration:** The computer guesses a flux profile, calculates the k_∞ distribution, solves the matrix, and repeats until convergence.

This allows us to model a reactor where control rods are halfway in, enrichment varies, and poisons are burning out non-uniformly.

7 The Microscopic View: Two-Group Flux in a Fuel Pin

So far today, we have applied Two-Group theory to the "global" reactor, treating the core as a homogeneous mixture of fuel and moderator. But what happens if we zoom in on a single 1 cm UO_2 fuel pellet surrounded by water?

By separating the neutrons into Fast and Thermal groups, we can finally explain the exact local flux shapes inside a heterogeneous core.

7.1 The Unit Cell Dynamics

Consider a cylindrical unit cell consisting of the fuel pellet, the cladding, and the surrounding water channel. The sources and sinks for our two groups are now physically separated:

- **Fission (Fast Source):** Happens *only* inside the fuel pellet.
- **Thermalization (Thermal Source):** Happens almost entirely in the water.
- **Absorption (Thermal Sink):** Happens heavily inside the fuel pellet.

7.2 The Local Flux Profiles

If we plot the two groups across this unit cell, they look like mirror images:

1. **The Fast Flux (ϕ_1):** Peaks in the dead center of the fuel pellet where the fissions are occurring. It drops off as it moves out into the water, because it is colliding and slowing down (being removed to Group 2).
2. **The Thermal Flux (ϕ_2):** Peaks out in the water where it is being created. As thermal neutrons diffuse *inward* toward the fuel, they are rapidly absorbed by the U-235 and U-238. This causes the thermal flux to plummet at the center of the pellet.

Nuclear engineers call this massive dip in the center the **Thermal Flux Depression**.

7.3 The Limit of Diffusion Theory

Earlier in the semester, we discussed why fuel pellets are sized at roughly 1 cm in diameter. If the pellet were much thicker, the thermal flux would depress so completely that the U-235 in the dead center would never see a thermal neutron—it would just be wasted dead weight!

A Critical Caveat: While the qualitative shapes we just described are correct, we cannot actually use the Two-Group *Diffusion* equations we derived today to calculate the exact math for this pin cell.

Why? Because Diffusion Theory (Fick's Law) assumes that the medium is uniform over distances of several **Mean Free Paths**. The mean free path of a thermal neutron in a fuel pellet is on the order of millimeters—which is exactly the scale of the pellet itself! On this microscopic scale, neutrons don't act like a diffusing gas; they act like individual bullets. To solve the pin cell accurately, reactor physicists must abandon Diffusion Theory and use the much more rigorous **Neutron Transport Equation** (which tracks exact angular trajectories).

Historical Note: The Limit of Diffusion & The Birth of Monte Carlo
We cannot actually use the Diffusion equations to calculate the exact math for this pin cell. Diffusion Theory assumes the medium is uniform over several Mean Free Paths. But the mean free

path of a thermal neutron in a fuel pellet is on the scale of the pellet itself! On this microscopic scale, neutrons act like individual bullets, requiring the rigorous **Neutron Transport Equation**.

During the Manhattan Project, Stanislaw Ulam (inspired by playing solitaire) and John von Neumann realized they could solve this by treating neutrons like bullets and "ray tracing" their paths using random probabilities. Von Neumann programmed the ENIAC computer to track thousands of simulated neutrons bouncing around a core, rolling virtual dice to determine scatters, absorptions, or fissions. They named this the **Monte Carlo method**. While it remains the gold standard for reactor physics computations today (e.g., the MCNP code), its impact extends far beyond nuclear engineering. Today, Monte Carlo algorithms are used globally to price financial derivatives on Wall Street, render photorealistic CGI in Hollywood, simulate protein folding in biology, and train machine learning models.

References

1. **Textbook:** Lamarsh, J.R. and Baratta, A.J., *Introduction to Nuclear Engineering*, 4th Edition. Section 6.7 (Multigroup Calculations).

2. **Visualization:** Wikipedia, *Neutron Reflector*.

3. **Historical Computing:** Metropolis, N., "The Beginning of the Monte Carlo Method," *Los Alamos Science* (1987). (A firsthand historical account)

Appendix: Deriving the Reflector Savings (δ)

In one-speed theory, deriving the reflector savings was relatively simple. In Two-Group theory, a rigorous exact solution requires matching four boundary conditions at the core-reflector interface ($x = a$):

1. Fast Flux continuity: $\phi_{1C}(a) = \phi_{1R}(a)$
2. Fast Current continuity: $D_{1C}\phi'_{1C}(a) = D_{1R}\phi'_{1R}(a)$
3. Thermal Flux continuity: $\phi_{2C}(a) = \phi_{2R}(a)$
4. Thermal Current continuity: $D_{2C}\phi'_{2C}(a) = D_{2R}\phi'_{2R}(a)$

Plugging the spatial solutions into these four conditions yields a 4×4 matrix of coefficients. For the reactor to be critical, the determinant of this matrix must be zero. This yields a messy transcendental equation that is solved via computer.

The Fast Leakage Approximation

However, we can derive a highly accurate, simple expression for δ by recognizing the physics of the problem: **most neutrons that leak out of a large core do so while they are Fast.** Therefore, the fast boundary conditions largely dictate the reflector savings.

Let's look only at the Fast group. In the core, assuming a large reactor where transient effects near the boundary are small, the fast flux follows the fundamental buckling cosine shape:

$$\phi_{1C}(x) = A\cos(Bx)$$

In the reflector, as we derived earlier, the fast flux is an exponential decay. Letting $z = x - a$:

$$\phi_{1R}(z) = Ce^{-\kappa_1 z}$$

Step 1: Match the Flux at the Boundary ($x = a$, or $z = 0$)

$$A\cos(Ba) = Ce^0 \implies A\cos(Ba) = C$$

Step 2: Match the Current at the Boundary Current is $J = -D\nabla\phi$. Taking the derivatives of our fluxes and equating them:

$$-D_{1C}(-AB\sin(Ba)) = -D_{1R}(-\kappa_1 Ce^0)$$

$$D_{1C}AB\sin(Ba) = D_{1R}\kappa_1 C$$

Step 3: Divide Current by Flux To eliminate the unknown constants A and C, we divide the current equation by the flux equation:

$$\frac{D_{1C}AB\sin(Ba)}{A\cos(Ba)} = \frac{D_{1R}\kappa_1 C}{C}$$

$$D_{1C}B\tan(Ba) = D_{1R}\kappa_1$$

Step 4: Introduce the Reflector Savings (δ) We define the extrapolated boundary of the core as $\tilde{a} = a + \delta$. By definition, the flux must go to zero at this extrapolated boundary, meaning $B\tilde{a} = \pi/2$. Therefore:

$$B(a + \delta) = \frac{\pi}{2} \implies Ba = \frac{\pi}{2} - B\delta$$

Substitute this back into our tangent term:

$$\tan(Ba) = \tan\left(\frac{\pi}{2} - B\delta\right)$$

Using trigonometric identities, $\tan(\pi/2 - \theta) = \cot(\theta)$:

$$\tan(Ba) = \cot(B\delta)$$

For a large commercial reactor, the buckling B is very small, making the term $B\delta$ very small. The small-angle approximation for cotangent is $\cot(\theta) \approx 1/\theta$. Thus:

$$\cot(B\delta) \approx \frac{1}{B\delta}$$

Step 5: Solve for δ Substitute this back into our equation from Step 3:

$$D_{1C}B\left(\frac{1}{B\delta}\right) \approx D_{1R}\kappa_1$$

The B terms cancel out brilliantly:

$$\frac{D_{1C}}{\delta} \approx D_{1R}\kappa_1$$

Solving for δ, and recalling that $\kappa_1 = 1/\sqrt{\tau_R}$:

$$\delta \approx \frac{D_{1C}}{D_{1R}\kappa_1} = \frac{D_{1C}}{D_{1R}}\sqrt{\tau_R}$$

The Physical Result: The reflector savings is directly proportional to the square root of the Fermi Age (τ_R) of the reflector material! A reflector that allows fast neutrons to travel further before thermalizing (larger τ) pushes the extrapolated boundary further out. Because water has a significant Fermi Age ($\tau \approx 27$ cm^2), it provides excellent reflector savings, shrinking the physical size of the core required for criticality.